

Statement of Work Requirements-24-2598

Renewal of Lidar System Maintenance, System Support, and Licensing NOAA/NOS/NGS/RSD

1. Introduction/Background

The National Oceanic and Atmospheric Administration (NOAA) is congressionally mandated to survey the National Shoreline and nearshore bathymetry. Currently, the National Geodetic Survey (NGS), Remote Sensing Division (RSD) utilizes a Leica Chiroptera-5 topographic/bathymetric lidar system to support the Coastal Mapping Program (CMP) safety of navigation mission to update NOAA Nautical Charts as well as:

- emergency response activities related to manmade and natural disasters
- airport obstruction surveying in support of the Aeronautical Survey Program (ASP)
- coastal change analysis, ecosystem mapping and monitoring
- storm surge modeling
- related research and development initiatives

This state-of-the-art system has provided enhanced mapping capabilities when compared to traditional aerial photographic imagery methods by providing cost-effective, digital elevation data with high spatial resolution and accuracy. This system is specifically designed for topographic and bathymetric use and data is merged with aerial photographic imagery taken simultaneously in order to create a seamless depiction of shoreline and terrain features. Ability to rapidly preview data for basic quality and completeness during and immediately after flight is important.

In order to provide tidal-datum-based shoreline and nearshore bathymetry as efficiently and effectively as possible for a wide variety of project sites, NOAA requires the capability to collect seamless, high-resolution elevation data on both sides of the land water interface, including in the marine nearshore zone to a minimum depth of 5 meters. To meet this need, an airborne topographic-bathymetric lidar system, capable of simultaneously mapping both the subaqueous and subaerial areas of the coastal zone is required. Additionally, topographic-bathymetric lidar data enable determination of water depths for application on NOAA Nautical Charts shoreward of the 4 meter / Navigation Limit Line (NALL) – an area that is traditionally difficult and dangerous to survey with marine vessels and therefore contains the greatest amount of gaps/uncertainty on charts. Other applications of topographic-bathymetric lidar with NOAA/NOS include coral reef habitat mapping, nearshore modeling including inundation and storm surge, and riverbed mapping. Simultaneously serving these various geospatial data needs will strengthen NOAA's Integrated Ocean and Coastal Mapping (IOCM) role, as well as meeting NOAA mission requirements.

2. Requirements

The government requires continued support services, maintenance, and licensing of a Leica Chiroptera-5 topographic-bathymetric lidar system and associated software.

The government intends to have one full year of Customer Care Package (CCP) Gold level support on

hardware and software maintenance, licensing, and support with four additional option years for CCP Gold level support on hardware and software maintenance, licensing, and support on the existing Leica Chiroptera-5 topographic-bathymetric lidar system and associated software. Maintenance and software updates shall be performed throughout the applicable performance period. The vendor shall have the ability to provide around-the-clock technical service and troubleshooting.

A Technical Service Representative shall be able to respond on-site or remote to the Aircraft Operations Center in Lakeland, FL, within 12 hours. When technical issues are encountered, the vendor shall make every effort to have them resolved within 24 hours. In cases where the issue cannot be quickly resolved on-site, the vendor shall provide an equal or better replacement (loaner) system or replace the failed component until technical issues with the owned system are resolved.

Interim, no-charge, substantially-equivalent product will be provided during maintenance and/or repair periods subject to equipment availability and logistics limitations.

DRAFT

3. Hardware Maintenance on following:

Chiroptera-5 System (S/N: 5937)

Sensor Head: Part no 2008930		
Art nr.	Description	Qty
973436	CH-5 Sensor unit	1
798528	CH82 Camera Head for Leica RCD30, 80MP, 4-band. (S/N: 82656)	1
785422	Leica NAG-D 4.0/50 lens (50mm) (S/N: 50204)	1
789322	Protective UV filter for NAT-D and NAG-D	1
835075	CNUS5-H LCI-100C, 500 Hz IMU, class 5 (S/N: 1296)	1

Sensor Control System: Part no 2009533		
Art nr.	Description	Qty
973438	Control System CH-5	1
872296	CC43 Camera Controller with SPAN and IMU interface (S/N: 43573)	1
757727	GNSS 1x4 L1/L2 splitter TNC	1
834747	GNSS antenna cable w. TNC female, 0,9 m	1
806037	IMU Cable 8 m	1
7000037	Cable Harness Leica 8m	1
7000036	Cable Harness Topo/Shallow 8m	2
846467	Leica 0.8m Camera Cable	1

Sensor Peripherals		
Art nr.	Description	Qty
799381	PD60 Pilot Display (S/N: 60416)	1
808321	OC60 Operator Console Display AHAB OC (S/N: 65483)	1
764189	PS56 Power Supply	1
821446	AC Power Cable US Wall Outlet Plug	2

Sensor Peripherals		
846863	Ground Power Supply Cable	1
752446	OC60/PD60 Accessories	1
989027	OC61 Operator Console WIN-10 lic (S/N: OC61045)	1
918803	OC60 Holder kit	1
805847	OC60/PD60 Interface Cable, 5m	2
805848	OC60/PD60 Interface Cable, 7m	1
846858	Safety Box	2
846859	Operator Safety Ctrl Cable	1
846860	Pilot Safety Ctrl Cable	1
743070	IS40 Interface Stand, hold OC60	1
846864	SSD 2TB	4
764192	Cable PD56/CC3x to PS56	1

PAV100		
Art nr.	Description	Quantity
805832	PAV100 gyro stabilized mount (S/N 18796)	1
784811	Cable CC3x to PAV80, 3m	1
798771	Power Cable PAV80 10A, 3m	1

4. Software Maintenance on following:

Leica LIDAR Survey Studio (LSS)	
Product	Serial Number
LIDAR Survey Studio for Chiroptera	235
LIDAR Survey Studio for Chiroptera	345
LIDAR Survey Studio for Chiroptera	247

Description of LSS software and capabilities:

- Point cloud generation for lidar
- Lidar calibration tool
- 3D viewing/editing tool
- Automatic water refraction correction
- Automatic data classification
- Waveform viewing
- Image viewing
- Lidar data Q/A tools
- Lidar data export

Leica HxMap/IPASCO Imagery Software	
Product	EID
HxMap RCD, SW lic	00107-11518-00031-50062-0EFFD
HxMap RCD, SW lic	0107-11520-00031-50062-7A3D1
HxMap Ingest, SW lic	00107-11519-00031-50062-60E96
HxMap Ingest, SW lic	00107-11521-00031-50062-8F63B
ORIMA ABS, software license	00107-11522-00031-50062-1E2E3
IPAS CO SW license	00107-11523-00031-50062-1B39A
HxMap Ortho Generator, SW lic	00107-35352-00032-01987-9A98F

Description of imagery software and capabilities:

- Imagery Development
- Imagery Orthorectification
- Boresight calibration
- Aerotriangulation
- Camera orientation calculation

Leica Flight Pro	
Product	Serial Number
FlightPro Flight Guidance, License	32246
FlightPro Remote Control, License	32246
FlightPro Sensor Control RCD, License	32246

Description of flight management system software and capabilities:

- Flight guidance and sensor control

Novatel Inertial Explorer	
Product	Serial Number
Inertial Explorer License	459
TerraStar NRT for IE Subscription (1 year)	n/a

Global Navigation Satellite System (GNSS) and Inertial Measurement Unit (IMU) software with the following capabilities:

- Able to control and configure GNSS/IMU hardware
- Able to post-process kinematic GNSS and IMU data
- 1 year Novatel TerraStar NRT Correction Services subscription

Leica Mission Pro	
Product	EID
MissionPro, Floating License	00107-11516-00031-50062-89BBE
MissionPro, Floating License	00107-11517-00031-50062-E3D50

Mission planning software with the following capabilities:

- Flight line planning for both Lidar and Imagery sensors

IT Security Requirements

The Assessment and Authorization (A&A) requirements of Clause 48 CFR 1352.239-72 do not apply, and a Security Accreditation Package is not required.

The following is a list of relevant information the Department of Commerce (DOC) Information Technology (IT) Security takes into consideration when assessing the security requirements and risk for procuring IT products and services.

- The Contractor shall only supply Information Technology (e.g., hardware, software, firmware, etc.) and/or product license and shall not require the use of any contractor owned equipment. The contractor will not have remote access to any government owned equipment. The Contractor shall ensure the acquiring Information

- Technology provided disables “phone home” or similar capabilities.
- b) Product support and supporting documentation allows for the sanitization of the acquiring Information Technology and/or product license upon the disposal of the product. Product shall allow the use one of the approved methods for secure destruction of data/information including unclassified but sensitive information. Approved methods include NIST Special Publications 800-88 *Guidelines for Media Sanitization* (<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-88r1.pdf>) or the National Security Agency in the *Media Destruction Guidance* (<https://www.nsa.gov/Resources/Media-Destruction-Guidance/>).
 - c) The acquiring information technology is engineered for trustworthy and security that is consistent with the engineering-based trustworthy secure solutions as outlined in NIST Special Publication 800-160 *Systems Security Engineering: Considerations for a Multidisciplinary Approach in the Engineering of Trustworthy Secure Systems* (<https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-160v1.pdf>).
 - d) The acquiring information technology complies with Office of Management and Budget Memorandum M-07-18 *Ensuring New Acquisitions Include Common Security Configurations* and the Federal Acquisitions Regulations (FAR) 39.101(d) regulations require NIST common security configuration checklists including United States Government Configuration Baseline (USGCB) initiative. More information is available at National Checklist Program (NCP), (<https://nvd.nist.gov/ncp/repository>), the U.S. government repository of publicly available security checklists (or benchmarks) that provide detailed low-level guidance on setting the security configuration of operating systems and applications. The acquiring information technology takes into account the following requirements:
 - i. Software procured meets the standard installation, operation, maintenance, updates and/or patching of software shall not alter the configuration settings from the approved USGCB/other secure configuration.
 - ii. Applications designed for normal end users run in the standard user context without elevated system administration privileges.
 - iii. Supporting documentation or reference(s) that describes the security capabilities, the design and development processes and the testing and evaluation procedures used by the product or services being provided for this acquisition.
 - iv. Supporting documentation or reference(s) that describes all product or service updates and enhancements as they are implemented. The product or service supporting documentation could be the user and system administrator guides, which is documents the functional properties of the security controls employed to permit the analysis and testing of the security controls.
 - e) The acquiring information technology complies with the Homeland Security Presidential Directive 12 (HSPD-12) *Policy for a Common Identification Standard for Federal Employees and Contractors* requirements from FAR 4.1302 stating: (a) In order to comply with Federal Information Processing Standard (FIPS) 201-3 *Personal Identity Verification (PIV) of Federal Employees and Contractors* (<https://doi.org/10.6028/NIST.FIPS.201-3>), agencies must purchase only approved personal identity verification products and services, (b) Agencies may acquire the approved products and services from the GSA, Federal Supply Schedule 70, Special Item Number (SIN) 132-62, HSPD-12 Product and Service Components, in accordance with ordering procedures outlined in FAR Subpart 8.4.
 - f) The acquiring information technology complies with Internet Protocol Version 6 (IPv6) requirements from FAR part 11.002 requirements that state the agency Chief Information Officer can waive this requirement provided the acquisition requirements documents the appropriate technical capabilities defined in the USGv6 Profile available in the NIST Special Publication (SP) 500-267 (<https://www.nist.gov/programs-projects/usgv6-program>) and the corresponding declarations of conformance. To meet this requirement each DOC acquisition of IP protocol technology must express requirements for IPv6 capabilities in terms of the USGv6 Profile (i.e., using the USGV6 Capabilities Check List) and vendors must be required to document their product’s support of the requested capabilities through the USGv6 test program (<https://www.nist.gov/programs-projects/usgv6-program/usgv6-revision-1>) using the USGv6 Suppliers Declaration of Conformity.